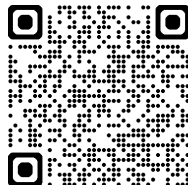


# Logical Theories and Logical Views

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# Aims

Logical  
Theories and  
Logical Views

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Logical  
Systems and  
Logical Views

Systems of  
Applications

Theories of  
Logic

- To explain what are logical theories and views, and how are they different from logical systems.
- To provide a way to visualise logical theories and views.
- To provide visual examples of logical theories and views.

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# Logical systems

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## Definition

A *system of logic* is a pair  $S = \langle \mathbf{Fo}_S, \mathbf{Cn}_S \rangle$ , where  $\mathbf{Fo}_S$  is a set of formulae and  $\mathbf{Cn}_S \subseteq 2^{\mathbf{Fo}_S} \times \mathbf{Fo}_S$  is a consequence relation.

## Examples

- classical sentential and first-order logics,
- modal logics M, B, S4, S5, etc.,
- Heyting's sentential intuitionistic logic, Jaskowski discussive logic, etc.

## Notation

' $(\mathbf{A}, F) \in \mathbf{Cn}_S$ ' means that  $F \in \mathbf{Fo}_S$  is an  $S$ -logical consequence (or  $S$ -consequence) of the set of formulae in  $\mathbf{A} \subseteq \mathbf{Fo}_S$  – i.e., it follows from  $\mathbf{A}$  *logically*, in the sense formalised by  $S$ .

# Logical theories and views

Cf. Hjortland, 'What counts as evidence for a logical theory?', 2019.

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## Remark

A *theory* of logic is more than a formal system of logic. A formal system, as any formalism, may be used for this or that purpose, context, topic, etc.

## Desideratum

A logical theory must be an account of what system or systems count as logical systems and their purposes or contexts or topics of *correct applicability*.

# The received view

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## Formulation

Many mathematical systems may be regarded as logical systems, but only one (*monism*) – standard aka classical first-order logic (*classicalism*) – is correct, and it applies to all topics (*universalism*) in the same way (*topic-neutralism*).

# Challenges to the received view

## Contra monism (pluralism)

'[W]e defend what we call *logical pluralism*, the view that there is more than one genuine deductive consequence relation'.

(Beall & Restall, *Logical Pluralism*, 2006, p. 3)

## Contra universalism

In wisdom is no logic.

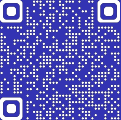
(Brower, 'De onbetrouwbaarheid der logische principes', 1905)

## Contra topic-neutrality (topic-specificism)

There is no single logic that is independent of any content, but in each domain a logic is adequate. The logical and the physical, the formal and the real, are interdependent.

(Destouches-Février, *La structure des théories physiques*, 1951, p. 88)





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# Theories of logic

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## Desiderata for the notion

We want a theory of logic to state:

- which mathematical systems are system of logics;
- what things qualify as logical topics (i.e., those topics about which logic is topic-neutral or not);
- how those systems of logic above are correctly applicable or not to these topics.

In order to define such a notion of *theory of logic*, I will first introduce the notion of a *system of applications*.

# System of applications: Definition

## Definition

A *system of applications* is a triple  $\mathfrak{A} = \langle \mathcal{T}_{\mathfrak{A}}, \mathcal{S}_{\mathfrak{A}}, \mathcal{R}_{\mathfrak{A}} \rangle$  such that:

- $\mathcal{T}_{\mathfrak{A}} = \{\mathbb{T}_1, \dots, \mathbb{T}_l\}$  is a non-empty set of topics; and
- $\mathcal{S}_{\mathfrak{A}} = \{S_1, \dots, S_m\}$  is a non-empty set of systems of logic;
- $\mathcal{R}_{\mathfrak{A}} = \{\mathbf{C}_{\mathfrak{A}}, \mathbf{R}_1, \dots, \mathbf{R}_n\}$  is a set of relations such that  $\mathbf{C}_{\mathfrak{A}}, \mathbf{R}_1, \dots, \mathbf{R}_n \subseteq \mathcal{T}_{\mathfrak{A}} \times \mathcal{S}_{\mathfrak{A}}$ , where  $\mathbf{C}_{\mathfrak{A}}$  is  $\mathfrak{A}$ 's distinguished relation.

## Pre-image and image under $\mathbf{R} \in \mathcal{R}_{\mathfrak{A}}$ and $\mathfrak{A}$

The *pre-image*  $\mathbf{Pr}_{\mathbf{R}}(\mathcal{S})$  of  $\mathcal{T} \subseteq \mathcal{T}_{\mathfrak{A}}$  and the *image*  $\mathbf{Im}_{\mathbf{R}}(\mathcal{T})$  of  $\mathcal{S} \subseteq \mathcal{S}_{\mathfrak{A}}$  under  $\mathbf{R} \in \mathcal{R}_{\mathfrak{A}}$  are defined as follows:

$$\mathbf{Pr}_{\mathbf{R}}(\mathcal{S}) \stackrel{\text{DF}}{=} \{\mathbb{T} \in \mathcal{T}_{\mathfrak{A}} : (\mathbb{T}, S) \in \mathbf{C}_{\mathfrak{A}}, \text{ for some } S \in \mathcal{S}\},$$

$$\mathbf{Im}_{\mathbf{R}}(\mathcal{T}) \stackrel{\text{DF}}{=} \{S \in \mathcal{S}_{\mathfrak{A}} : (\mathbb{T}, S) \in \mathbf{C}_{\mathfrak{A}}, \text{ for some } \mathbb{T} \in \mathcal{T}\},$$

$$\mathbf{Pr}_{\mathfrak{A}}(\mathcal{S}) \stackrel{\text{DF}}{=} \mathbf{Pr}_{\mathbf{C}_{\mathfrak{A}}}(\mathcal{S}), \quad \mathbf{Im}_{\mathfrak{A}}(\mathcal{T}) \stackrel{\text{DF}}{=} \mathbf{Im}_{\mathbf{C}_{\mathfrak{A}}}(\mathcal{T}).$$

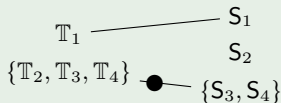
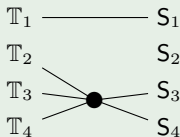
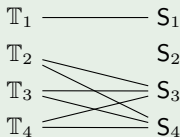
# System of applications: Diagrams

## Explanation

A relation  $\mathbf{R} \in \mathcal{R}_{\mathfrak{A}}$  may be represented diagrammatically as a two-column graph, with the left column comprising topics, the *topics column*, and the right column comprising logical systems, *logics* or *systems column*.

## Example 1

Consider the relation:  $\mathbf{R} = \{\langle \mathbb{T}_1, S_1 \rangle, \langle \mathbb{T}_2, S_3 \rangle, \langle \mathbb{T}_2, S_4 \rangle, \langle \mathbb{T}_3, S_3 \rangle, \langle \mathbb{T}_3, S_4 \rangle, \langle \mathbb{T}_4, S_3 \rangle, \langle \mathbb{T}_4, S_4 \rangle\}$ . In words,  $\mathbb{T}_1$  and  $S_1$  are  $\mathbf{R}$ -related exclusively, while all of  $\mathbb{T}_2, \mathbb{T}_3, \mathbb{T}_4$  are all related to all of  $S_3, S_4$ . The following diagrams are representations of  $\mathbf{R}$ :



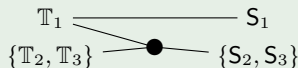
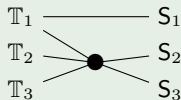
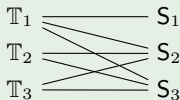
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## Example 2

Consider the relation:  $\mathbf{R} = \{\langle \mathbb{T}_1, S_1 \rangle\} \cup (\{\mathbb{T}_1, \mathbb{T}_2, \mathbb{T}_3\} \times \{S_2, S_3\})$ . In words,  $\mathbb{T}_1$  and  $S_1$  are  $\mathbf{R}$ -related exclusively, while all of  $\mathbb{T}_2, \mathbb{T}_3, \mathbb{T}_4$  are all related to all of  $S_3, S_4$ . The following diagrams are representations of  $\mathbf{R}$ :



# System of applications: Diagrams

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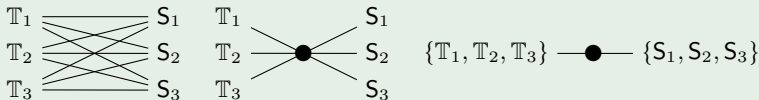
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## Explanation

A relation  $\mathbf{R} \in \mathcal{R}_{\mathcal{A}}$  may be represented diagrammatically as a two-column graph, with the left column comprising topics, the *topics column*, and the right column comprising logical systems, *logics* or *systems column*.

## Example 3

Consider the relation:  $\mathbf{R} = \{\mathbb{T}_1, \mathbb{T}_2, \mathbb{T}_3\} \times \{\mathbb{S}_1, \mathbb{S}_2, \mathbb{S}_3\}$  In words,  $\mathbb{T}_1$  and  $\mathbb{S}_1$  are  $\mathbf{R}$ -related exclusively, while all of  $\mathbb{T}_2, \mathbb{T}_3, \mathbb{T}_4$  are all related to all of  $\mathbb{S}_3, \mathbb{S}_4$ . The following diagrams are representations of  $\mathbf{R}$ :



# Classes of systems of applications

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## Classes of systems of applications

- We can speak about a class of systems of applications by specifying some properties which are satisfied only by the members of that class.
- For instance, we might speak of all systems of applications  $\mathfrak{A}$  such that  $\mathcal{T}_{\mathfrak{A}}$  and  $\mathcal{S}_{\mathfrak{A}}$  are singletons, and  $\mathbf{C}_{\mathfrak{A}} = \mathcal{T}_{\mathfrak{A}} \times \mathcal{S}_{\mathfrak{A}}$  (the only topic is related to the only system of logic).

## Schematic diagrams

- However, in order to represent one such system of applications diagrammatically, we cannot use specific diagrams, but schematic diagrams.
- In a schematic diagram, the terms for denoting topics and systems are variables, unless otherwise specified.

# Schematic diagrams: Notation 1

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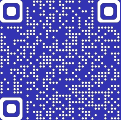
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| Statement                                                                                      | Representation                                           |
|------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| $\exists T \in \mathcal{T} \exists S \in \mathcal{S} (\langle T, S \rangle \in \mathbf{R})$    | $\mathcal{T} \text{ ————— } \mathcal{S}$                 |
| $\exists T \in \mathcal{T} \forall S \in \mathcal{S} (\langle T, S \rangle \in \mathbf{R})$    | $\mathcal{T} \text{ — } \bigcirc \text{ — } \mathcal{S}$ |
| $\exists S \in \mathcal{S} \forall T \in \mathcal{T} (\langle T, S \rangle \in \mathbf{R})$    | $\mathcal{T} \text{ — } \bigcirc \text{ — } \mathcal{S}$ |
| $\forall T \in \mathcal{T} \exists S \in \mathcal{S} (\langle T, S \rangle \in \mathbf{R})$    | $\mathcal{T} \text{ — } \bullet \text{ — } \mathcal{S}$  |
| $\forall S \in \mathcal{S} \exists T \in \mathcal{T} (\langle T, S \rangle \in \mathbf{R})$    | $\mathcal{T} \text{ — } \bullet \text{ — } \mathcal{S}$  |
| $\forall T \in \mathcal{T} \forall S \in \mathcal{S} (\langle T, S \rangle \in \mathbf{R})$    | $\mathcal{T} \text{ — } \bullet \text{ — } \mathcal{S}$  |
| $\exists T \in \mathcal{T} \exists S \in \mathcal{S} (\langle T, S \rangle \notin \mathbf{R})$ | $\mathcal{T} \text{ — } + \text{ — } \mathcal{S}$        |
| $\exists T \in \mathcal{T} \forall S \in \mathcal{S} (\langle T, S \rangle \notin \mathbf{R})$ | $\mathcal{T} \text{ — } \bigcirc \text{ — } \mathcal{S}$ |
| $\exists S \in \mathcal{S} \forall T \in \mathcal{T} (\langle T, S \rangle \notin \mathbf{R})$ | $\mathcal{T} \text{ — } \bigcirc \text{ — } \mathcal{S}$ |
| $\forall T \in \mathcal{T} \exists S \in \mathcal{S} (\langle T, S \rangle \notin \mathbf{R})$ | $\mathcal{T} \text{ — } \bullet \text{ — } \mathcal{S}$  |
| $\forall S \in \mathcal{S} \exists T \in \mathcal{T} (\langle T, S \rangle \notin \mathbf{R})$ | $\mathcal{T} \text{ — } \bullet \text{ — } \mathcal{S}$  |
| $\forall T \in \mathcal{T} \forall S \in \mathcal{S} (\langle T, S \rangle \notin \mathbf{R})$ | $\mathcal{T} \text{ — } \bullet \text{ — } \mathcal{S}$  |



# Schematic diagrams: Notation 2

| Statement                                                                        | Repr.                     | Statement                                                                        | Repr.                     |
|----------------------------------------------------------------------------------|---------------------------|----------------------------------------------------------------------------------|---------------------------|
| $\exists_{T \in \mathcal{T}} \exists_S (\langle T, S \rangle \in \mathbf{R})$    | $\mathcal{T} \text{ —}$   | $\exists_T \exists_{S \in \mathcal{S}} (\langle T, S \rangle \in \mathbf{R})$    | $\text{— } \mathcal{S}$   |
| $\exists_{T \in \mathcal{T}} \forall_S (\langle T, S \rangle \in \mathbf{R})$    | $\mathcal{T} \oplus$      | $\exists_T \forall_{S \in \mathcal{S}} (\langle T, S \rangle \in \mathbf{R})$    | $\oplus \mathcal{S}$      |
| $\exists_S \forall_{T \in \mathcal{T}} (\langle T, S \rangle \in \mathbf{R})$    | $\mathcal{T} \oplus$      | $\exists_{S \in \mathcal{S}} \forall_T (\langle T, S \rangle \in \mathbf{R})$    | $\oplus \mathcal{S}$      |
| $\forall_{T \in \mathcal{T}} \exists_S (\langle T, S \rangle \in \mathbf{R})$    | $\mathcal{T} \leftarrow$  | $\forall_T \exists_{S \in \mathcal{S}} (\langle T, S \rangle \in \mathbf{R})$    | $\leftarrow \mathcal{S}$  |
| $\forall_S \exists_{T \in \mathcal{T}} (\langle T, S \rangle \in \mathbf{R})$    | $\mathcal{T} \rightarrow$ | $\forall_{S \in \mathcal{S}} \exists_T (\langle T, S \rangle \in \mathbf{R})$    | $\rightarrow \mathcal{S}$ |
| $\forall_{T \in \mathcal{T}} \forall_S (\langle T, S \rangle \in \mathbf{R})$    | $\mathcal{T} \bullet$     | $\forall_T \forall_{S \in \mathcal{S}} (\langle T, S \rangle \in \mathbf{R})$    | $\bullet \mathcal{S}$     |
| $\exists_{T \in \mathcal{T}} \exists_S (\langle T, S \rangle \notin \mathbf{R})$ | $\mathcal{T} \mid$        | $\exists_T \exists_{S \in \mathcal{S}} (\langle T, S \rangle \notin \mathbf{R})$ | $\mid \mathcal{S}$        |
| $\exists_{T \in \mathcal{T}} \forall_S (\langle T, S \rangle \notin \mathbf{R})$ | $\mathcal{T} \downarrow$  | $\exists_T \forall_{S \in \mathcal{S}} (\langle T, S \rangle \notin \mathbf{R})$ | $\downarrow \mathcal{S}$  |
| $\forall_{T \in \mathcal{T}} \forall_S (\langle T, S \rangle \notin \mathbf{R})$ | $\mathcal{T} \bullet$     | $\forall_T \forall_{S \in \mathcal{S}} (\langle T, S \rangle \notin \mathbf{R})$ | $\bullet \mathcal{S}$     |



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# Complete theory of logic: Definition

## Definition

A (*complete*) *theory of logic* is a collection of statements stating all the following: (i) exactly which logical topics exist; (ii) exactly which mathematical systems qualify as logical systems; (iii) exactly which logical systems are correctly applicable to which logical topics.

## Definition: Theory of logic based on a system of applications

A theory of logic *is based on* a system of applications  $\mathfrak{A} = \langle \mathcal{T}_{\mathfrak{A}}, \mathcal{S}_{\mathfrak{A}}, \{\mathbf{C}_{\mathfrak{A}}\} \rangle$  iff the theory's statements is the classical logical closure of the following statements:

- $\mathbb{T}$  is a topic iff  $\mathbb{T} \in \mathcal{T}_{\mathfrak{A}}$ .
- $S$  is a system of logic iff  $S \in \mathcal{S}_{\mathfrak{A}}$ .
- $S$  is correctly applicable to  $\mathbb{T}$  iff  $\langle \mathbb{T}, S \rangle \in \mathbf{C}_{\mathfrak{A}}$ .

# Example 1: Generalistic classicalism

## Definition

*Generalistic classicalism* is the theory (of logic) based the system of applications  $\mathfrak{A} = \langle \mathcal{I}_{\mathfrak{A}}, \mathcal{S}_{\mathfrak{A}}, \{\mathbf{C}_{\mathfrak{A}}\} \rangle$ , where:

- $\mathcal{I}_{\mathfrak{A}} = \{\mathbb{G}\}$ , where  $\mathbb{G}$  is the most general topic;
- $\mathcal{S}_{\mathfrak{A}} = \{\mathbf{C}\}$ , where  $\mathbf{C}$  is classical logic; and
- $\mathbf{C}_{\mathfrak{A}} = \{\langle \mathbb{G}, \mathbf{C} \rangle\}$ , that is,

$$\mathbb{G} \text{ ————— } \mathbf{C}$$

# Example 2: Destouches-Fevrier conception

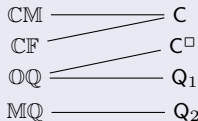
CM: classical mechanics; CF: classical field theory; OQ: old quantum mechanics;  
MQ: modern quantum mechanics;  $C^\square$ : an extension of C;  $Q_{1,2}$ : two quantum logics.

*[As] a deductive theory is constructed to adequately describe a domain of science, and ... a logic must constitute the rules of reasoning of such a theory, then this logic is not independent of the content of this theory, of its domain of adequacy. ... There is no single logic that is independent of any content, but in each domain a logic is adequate. The logical and the physical, the formal and the real, are interdependent. (P. Destouches-Fevrier, La structure des théories physiques, 1951, p. 88.)*

## Definition

D.-F.'s view is the theory (of logic) based the system of applications  $\mathfrak{A} = \langle \mathcal{T}_{\mathfrak{A}}, \mathcal{S}_{\mathfrak{A}}, \{C_{\mathfrak{A}}\} \rangle$ , where:

- $\mathcal{T}_{\mathfrak{A}} = \{\text{CM}, \text{CF}, \text{OQ}, \text{MQ}\};$
- $\mathcal{S}_{\mathfrak{A}} = \{\text{C}, C^\square, Q_1, Q_2\};$  and
- $C_{\mathfrak{A}} = \{\langle \text{CM}, \text{C} \rangle, \langle \text{CF}, \text{C} \rangle, \langle \text{OQ}, Q_1 \rangle, \langle \text{OQ}, C^\square \rangle, \langle \text{MQ}, Q_2 \rangle\}$ , that is,



*Thanks!*

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